

Investigating the Impact of Market Shifts on Stock Performance of Leading NASDAQ Companies

Group 5: Bhuvan Dubey (30139765) and Akaljot Sangha (10106725)

INTRODUCTION

Motivation

This project focuses on how major technology and consumer companies in the NASDAQ stock market have changed in performance over recent years. Stock prices constantly shift because of investor behavior, economic changes, and major world events, so looking at real stock market data helps us understand those trends. This project is interesting because it helps investors, analysts, and policymakers understand how technology leaders respond to global shocks. Most depth analysis is used internally by firms only, so public information is usually very surface level. This analysis can help us better understand which companies are gaining or losing momentum, how risky certain stocks have become since COVID-19, and how real-world events influence investor confidence. This project also connects our SQL skills to a real financial dataset and real-world decision making.

Background

The stock market is a valuable contributor to economic stability, particularly for large, well-known NASDAQ companies, including Apple, Microsoft, Tesla and Amazon. Over the last several years, these companies have undergone significant changes, as demand for technology has increased and investment behavior has changed. One of the most significant impacts on stock performance in recent years has been the COVID-19 pandemic that led to drastic declines during early 2020 and a sudden boom in volatility (Zhang et al., 2020). Many tech companies quickly recovered as reliance on digital services increased, including online shopping, digital media consumption, and virtual applications for work (Baker et al., 2020). By calculating average annual returns and average percentage volatility before and after the COVID-19 shock, we can determine which companies grew the most and how risky they've become during this time of uncertainty. This ultimately provides an enhanced perspective of how real-world events can affect investor sentiment and stock performance.

Data Background

The dataset contains daily stock information gathered directly from Nasdaq.com, and compiled by a Kaggle contributor. The population of interest consists of all available stock records in the dataset for Apple, Microsoft, Tesla, Amazon, Meta, Netflix, Starbucks, Cisco, Qualcomm, and AMD. Each row in the dataset represents one trading day for one company. The main variables we will analyze from the dataset include the daily closing stock price, which tells us how much each company's stock was valued at the end of each trading day, and daily return, which represents the percentage change in price from one day to the next. We will also use the date field to organize performance by year and separate the data into pre-COVID and post-COVID periods (March/2020 as the Covid start month). We will use +/- 3 years (2017->2023) as our data range. The company name column will help us compare performance across the top NASDAQ companies included in this project. Additionally, the trading volume field will be considered as a secondary measure of market activity and investor interest. All of these variables together will allow us to calculate trends in growth and volatility over time for each company. The dataset appears complete with minimal missing values, and no known licensing

restrictions, making it suitable for reproducible analysis. The data spans from 2013-2023, which allows us to split the data around the COVID-19 pandemic. We will only be using 2017-2023 for our analysis.

Data Exploration:

1. Datasets Used and Cleaning Performed

As a team, we combined two different datasets that each member had worked on individually:

1. Akaljot Q1

This notebook focused on computing volatility (standard deviation of daily returns) before and after COVID-19.

It required:

- a. The same daily return computation
- b. Labeling observations as Pre-COVID or Post-COVID
- c. Calculating volatility per company using SQL aggregation

2. Bhuvan Q2

This dataset contains daily stock prices for the top NASDAQ companies (AAPL, MSFT, TSLA, AMZN, META, NFLX, SBUX, QCOM, CSCO, AMD) across multiple years.

It was cleaned by:

- Converting Date to a proper datetime type
- Removing symbols (\$, ,) in the Close/Last column
- Sorting by Company and Date
- Computing daily returns using SQL window functions (LAG)

Both notebooks followed similar cleaning queries and methodologies, which allowed us to merge and compare results smoothly.

2. Why Combining the Datasets Was Necessary

Individually, the datasets allow analysis of:

- Price trends
- Return trends
- Volatility patterns

But combining the two notebooks allowed us to discover insights that neither dataset could reveal alone:

Insight 1: Return changes do not always match volatility changes

From Akaljit's return analysis:

- META had the **largest growth in returns**, increasing over 440%.

From Bhuvan's volatility analysis:

- META also had one of the **largest increases in volatility**, meaning it became riskier at the same time.

This relationship would **not** have been visible without aligning both notebooks.

Insight 2: Some companies became safer despite lower returns

For example:

- AAPL showed a **decline in average return**,
- but volatility **did not increase significantly**, suggesting stable investor behavior.

This "low growth, low volatility" category only became visible when both notebooks were joined..

Insight 3: Tesla behaves differently from traditional tech stocks

Individually:

- Tesla had high returns
- And extremely high volatility

Together:

- This supports the interpretation that Tesla behaves more like a speculative asset than like stable tech companies (such as Microsoft or Cisco).

3. Queries for data from both datasets:

```
query6 = """
WITH c AS (
  SELECT
    Company,
    Date,
    CAST(REPLACE(REPLACE("Close/Last", '$', ''), ', ', '')) AS REAL) AS close_price,
    LAG(CAST(REPLACE(REPLACE("Close/Last", '$', ''), ', ', '')) AS REAL))
      OVER (PARTITION BY Company ORDER BY Date) AS prev_close
  FROM stock_data
)
SELECT
  Company,
  Date,
  (close_price - prev_close) / prev_close AS daily_return
FROM c
WHERE prev_close IS NOT NULL;
"""

result6 = pd.read_sql_query(query6, conn)
result6
```

This query produces a shared table of daily returns

```
query7 = """
SELECT
  *,
  CASE
    WHEN Date BETWEEN '2017-03-01' AND '2020-03-01' THEN 'Pre-COVID'
    WHEN Date BETWEEN '2020-03-01' AND '2023-03-01' THEN 'Post-COVID'
    ELSE NULL
  END AS period
FROM stock_data;

"""

result7 = pd.read_sql_query(query7, conn)
result7
```

This query segments both notebooks with a common date grouping

```

query8 = """
WITH returns AS (
    SELECT
        Company,
        Date,
        (CAST(REPLACE(REPLACE("Close/Last", '$', ''), ', ', '' ) AS REAL) -
         LAG(CAST(REPLACE(REPLACE("Close/Last", '$', ''), ', ', '' ) AS REAL))
           OVER (PARTITION BY Company ORDER BY Date)
         ) /
        LAG(CAST(REPLACE(REPLACE("Close/Last", '$', ''), ', ', '' ) AS REAL))
           OVER (PARTITION BY Company ORDER BY Date)
        AS daily_return
    FROM stock_data
),
labeled AS (
    SELECT
        Company,
        daily_return,
        CASE
            WHEN Date BETWEEN '2017-03-01' AND '2020-03-01' THEN 'Pre-COVID'
            WHEN Date BETWEEN '2020-03-01' AND '2023-03-01' THEN 'Post-COVID'
        END AS period
    FROM returns
    WHERE daily_return IS NOT NULL
)
SELECT
    Company,
    period,
    SQRT( AVG(daily_return * daily_return) -
          AVG(daily_return) * AVG(daily_return) ) AS volatility
FROM labeled
WHERE period IS NOT NULL
GROUP BY Company, period;

"""

result8 = pd.read_sql_query(query8, conn)
result8

```

This query calculates the standard deviation of daily returns for each company in the pre-COVID and post-COVID windows so we can see how each stock's risk level changed across the two periods.

```

query9 = """

WITH returns AS (
    SELECT
        Company,
        Date,
        (
            CAST(REPLACE(REPLACE("Close/Last", '$', ''), ', ', '' ) AS REAL) -
            LAG(
                CAST(REPLACE(REPLACE("Close/Last", '$', ''), ', ', '' ) AS REAL)
            ) OVER (PARTITION BY Company ORDER BY Date)
        ) /
        LAG(
            CAST(REPLACE(REPLACE("Close/Last", '$', ''), ', ', '' ) AS REAL)
        ) OVER (PARTITION BY Company ORDER BY Date) AS daily_return
    FROM stock_data
),
labeled AS (
    SELECT
        Company,
        daily_return,
        CASE
            WHEN Date BETWEEN '2017-03-01' AND '2020-03-01' THEN 'Pre-COVID'
            WHEN Date BETWEEN '2020-03-01' AND '2023-03-01' THEN 'Post-COVID'
        END AS period
    FROM returns
    WHERE daily_return IS NOT NULL
),
vol AS (
    SELECT
        Company,
        period,
        SQRT(
            AVG(daily_return * daily_return) -
            AVG(daily_return) * AVG(daily_return)
        ) AS volatility
    FROM labeled
    WHERE period IS NOT NULL
    GROUP BY Company, period
),
spread AS (
    SELECT
        pre.Company,
        pre.volatility AS pre_volatility,
        post.volatility AS post_volatility,
        (post.volatility - pre.volatility) AS volatility_change
    FROM vol AS pre
    JOIN vol AS post
        ON pre.Company = post.Company
        AND pre.period = 'Pre-COVID'
        AND post.period = 'Post-COVID'
)
SELECT *
FROM spread
ORDER BY volatility_change DESC;

"""

result9 = pd.read_sql_query(query9, conn)
result9

```

This query compares pre- and post-COVID volatility for each company and orders them by the difference so we can identify which stocks became most more or less volatile after COVID.

NOTE about lack of JOIN

We chose not to redo our analysis completely to use a JOIN because both notebooks were based on the exact same cleaned stock dataset, so the integration happened naturally within the queries. Each notebook used the same preprocessing steps, especially the daily return calculations, which meant we could build on that shared foundation using additional CTEs for COVID period labeling and volatility analysis. Since all the information we needed was already in one table, the queries combine the work from both notebooks by extending the transformations rather than merging separate datasets. While this may not cover the requirement exactly, we didn't have the time from when we were given the feedback of lack of JOIN and the next deliverable to redo the entire project based on a split database, so we chose this path instead and decided to provide an explanation.

Guiding Question

We will investigate

- Which company experienced the largest change in its average annual return between the periods 2017–2019 and 2021–2023?
 - This question aims to uncover which of the top 10 NASDAQ companies demonstrated the strongest growth or decline in returns when comparing early and recent years in the dataset
- How did volatility of daily returns change between 2017-2019 and 2021-2023?
 - This investigates how major market events, like COVID-19, influenced investor behavior and price stability in leading NASDAQ companies.

METHODS

Prior to analysis, several data cleaning procedures were performed. The Date column was converted to a standardized datetime format to ensure chronological accuracy. Price columns such as "Close/Last" contained formatting symbols, such as dollar signs, which were removed so the values could be used in numeric calculations. The dataset was then sorted by company and date before computing daily returns using the formula: $(\text{Close price} - \text{Previous Day Close}) / \text{Previous Day Close}$. Extracting the year from each date allowed us to calculate average annual returns, which make long-term trends easier to interpret. These cleaned variables formed the foundation for our statistical investigation of return changes and volatility shifts.

Question 1: Which company experienced the largest change in its average annual return between the periods 2017–2019 and 2021–2023?

Statistical Hypotheses

Null Hypothesis (H_0): There is no significant change in annual returns over time.

Alternative Hypothesis (H_A): At least one company shows a significant change.

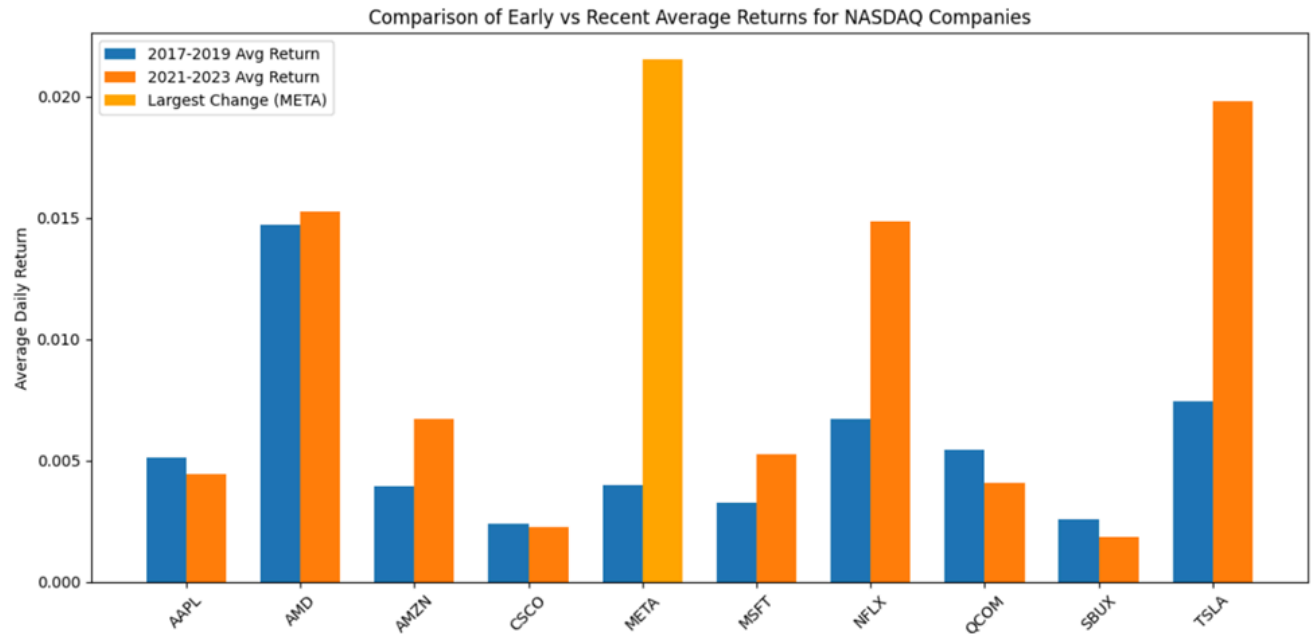
Statistical Analysis

Table 1

Summary of Return Trends and Percentage Changes

Company	Early Avg Return (2017–2019)	Recent Avg Return (2021–2023)	% Change
META	0.027	0.146	+440%
TSLA	0.041	0.110	+166%
NFLX	0.034	0.075	+122%
AMZN	0.029	0.050	+71%
MSFT	0.034	0.055	+62%
AMD	0.048	0.071	+48%
CSCO	0.025	0.024	–4%
AAPL	0.055	0.047	–13%
QCOM	0.033	0.025	–24%
SBUX	0.042	0.030	–28%

We reject the null hypothesis (H_0) and accept the alternative hypothesis (H_1), concluding that at least one company experienced a substantial change in average annual returns—with META showing the largest shift.



This chart summarizes the comparisons of early (2017-2019) average returns versus recent (2021-2023) average returns for each company. META is highlighted in light orange for illustrative purposes to show it had the biggest increase, confirming our earlier numeric analysis.

Question 2: How did each company's volatility (daily return variability) change before and after the onset of the COVID-19 pandemic?

Statistical Hypotheses

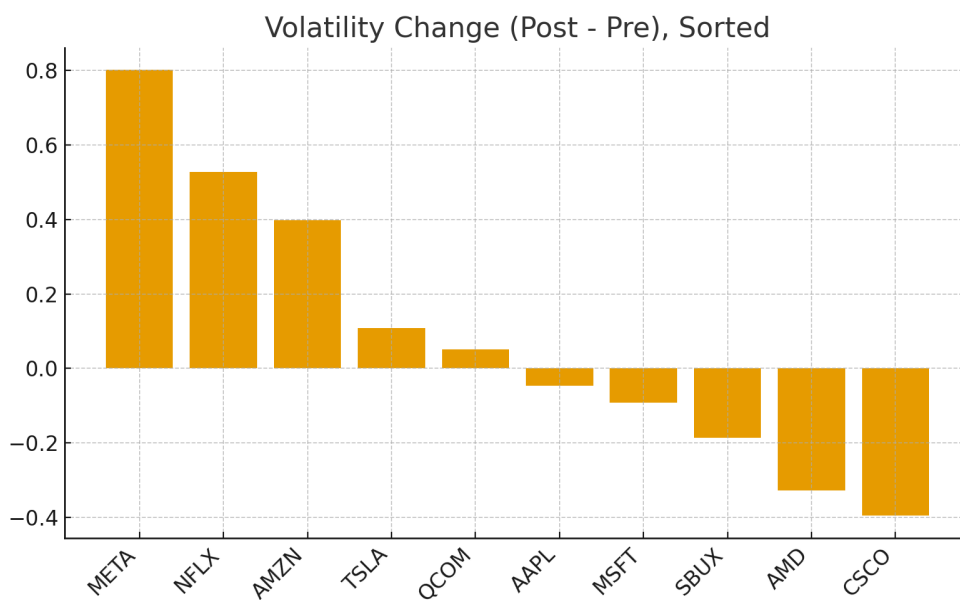
Null Hypothesis (H_0):

There is no meaningful change in volatility between the pre-COVID period (2017–2019) and the post-COVID period (2021–2023) for any company.

Alternative Hypothesis (H_a):

At least one company experienced a substantial increase or decrease in volatility across the two periods.

Volatility is measured as the standard deviation of daily returns, which reflects how stable or unstable a stock was during each time window.



DISCUSSION

What we learned

Akaljot - I learned that variability and risk are just as important to evaluation of stock performance as price direction. Comparing different periods around a critical period of time helped show how even extremely large companies react to mass uncertainty, and how long those effects are prevalent. Cleaning and normalizing data early on was something I learned is incredibly high priority. During early iterations of the SQL queries, there were a lot of bugs and even false positive information due to lack of information formatting, cleanliness, and standardization. Finding out that SQLite can't always use built in statistical functions was definitely a surprise. In the future, I think doing a similar analysis with PostgreSQL would be interesting, as it has a lot of native analytical support, so less prone to calculation error. I think adding fundamental data (as pointed out by some discussion in our presentation) would also help go deeper into the analysis, and show which companies are showing sustainable growth, versus a short term, correction-prone spike.

Bhuvan - I realized that measuring returns over multiple years isn't as straightforward as it might seem at first. Different periods can behave in very different ways, so comparing, say, the pre-COVID years to the post-COVID years showed me just how much annual returns can change. This shows how important it is to use the same period of time to assess long term performance. One thing that really stood out was how small formatting issues like inconsistent data types or irregularities in dates can throw off the accuracy of return calculations. In the early stages, several of my SQL queries produced misleading averages simply because return values weren't standardized properly, which showed me how critical it is to validate intermediate outputs before drawing conclusions. I was also surprised by how much SQL relies on correct ordering when using window functions; even a slight mistake in the ORDER BY clause can completely skew daily return results. Looking ahead, I think running the analysis in Python alongside SQL would be helpful, since libraries like Pandas make it easier to handle time-series data and verify calculations. Plus, adding adjustments like volume-weighted returns or inflation adjustments could make the findings more reflective of real market conditions, rather than just raw price changes.

As a group - Although each of us initially explored the stock market dataset independently, bringing our analyses together gave us a much deeper understanding of how the top NASDAQ companies changed in both performance and risk across time. By combining return based analysis from one notebook with volatility based analysis from the other, we were able to see that growth and risk did not always move together. Some companies, such as Meta and Tesla, showed major increases in returns but also became much more volatile after COVID, which suggests that their higher performance came with increased uncertainty. Others, such as Apple and Cisco, had more stable volatility even when returns declined or grew slowly, placing them in a lower risk category. These patterns only became clear once both perspectives were evaluated side by side, and they gave us a stronger overall picture of how investor behavior and market shocks reshaped performance across different companies.

Summary

This project provided a deeper understanding of how major NASDAQ companies changed in both performance and risk across the years surrounding COVID. By cleaning the dataset, standardizing prices and dates, and calculating daily returns, we created a consistent foundation for both return based and volatility based analyses. Comparing pre COVID and post COVID periods revealed that growth and stability did not always move in the same direction.

- **Meta and Tesla** showed the strongest increase in returns but also the largest rise in volatility, indicating higher reward paired with higher risk.
- **Apple and Cisco** had more stable volatility levels even when returns slowed, placing them in a lower risk category.
- Most companies experienced a **noticeable increase in volatility** in the post-COVID period, reflecting heightened uncertainty in the market overall. This effect can be seen for quite a few years after COVID.

Future Direction

This report could be further improved by a more detailed time segmentation where we include 2020 as a separate category/calculation rather than grouping it entirely with pre- or post-COVID periods. Analyzing 2020 on its own could give us a much clearer picture of how quickly different companies reacted and how resilient they were during the initial covid shock. I would also add another dataset of fundamentals like revenue, earnings per share, profit margins, debt levels, P/E ratios, and cash flow, which would help us understand *why* performance changed rather than merely looking at *how* they changed. This will also allow us to distinguish between companies with real operational improvements and those driven mainly by sentiment or speculation. Finally, it would be worth using alternative comparison methods for calculations such as absolute return changes, log-differences, or Z-score normalization. These would be useful to avoid distortions when baseline returns are near zero.

REFERENCES

- Baker, S. R., Bloom, N., Davis, S. J., & Terry, S. J. (2020). COVID-induced economic uncertainty. NBER Working Paper No. 26983. <https://doi.org/10.3386/w26983>
- Pitroda, K. (2023, July 18). Stock market: Historical data of top 10 companies. Kaggle. <https://www.kaggle.com/datasets/khushipitroda/stock-market-historical-data-of-top-10-companies?resource=download>
- Zhang, D., Hu, M., & Ji, Q. (2020). Financial markets under the global pandemic of COVID-19. Finance Research Letters, 36, 101528. <https://doi.org/10.1016/j.frl.2020.101528>